

Experiment No.	2
Experiment Title	Application of Naïve Bayes & K-Nearest Neighbors for Email Spam Classification ¹
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Anaconda Activation Command:

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source /opt/anaconda3/bin/activate anaconda-navigation
```

1 Objective

For this experiment I worked with Naïve Bayes (Gaussian and Multinomial) and K-Nearest Neighbors on the Spambase dataset, mainly to see how a probabilistic classifier stacks up against a distance-based one when the job is telling spam emails from ham.

2 Problem Statement

Each email in this dataset is represented as a set of frequencies – how often certain words and characters show up. Given that, the goal is simply to build something that can look at these frequencies and decide, reasonably reliably, whether a given email is spam.

3 Dataset Description

Dataset Name	Spambase Dataset
Dataset Source	Kaggle / UCI Machine Learning Repository
Number of Samples	4601
Number of Features	57
Number of Classes	2 (Spam, Ham)
Missing Values	None
Train-Test Split	70% Train, 30% Test

4 Exploratory Data Analysis

I used an automated EDA routine that puts together a 12-subplot summary of how the features are distributed and how they relate to each other. The lab spec called for Times New Roman at 15pt with bold fonts, so the plot was set up to follow that.

5 Data Preprocessing

KNN works by measuring Euclidean distance, so if one feature happens to have a much larger range than the rest, it ends up dominating that distance calculation whether it deserves to or not. `StandardScaler` takes care of this by putting all the numerical

features on the same footing before they go into the KNN model. Multinomial Naïve Bayes is a bit different – it needs non-negative values to work correctly, so I left those features in their original, unscaled form for that model. Beyond that, the data was split 70/30 into training and test sets.

6 Machine Learning Algorithm

Two models went into this experiment:

- **Naïve Bayes:** built on Bayes' theorem, with the (fairly strong) assumption that all features are independent of one another given the class. I used the multinomial variant since it handles count-style features like word frequencies well.
- **K-Nearest Neighbors (KNN):** doesn't really "learn" a model in the usual sense – it just looks at the K closest points to a new sample and goes with whichever class is most common among them, using Euclidean distance to decide "closest."

7 Experimental Setup

Python Version	3.10+
Scikit-Learn Version	1.0+
IDE	VS Code
Operating System	Linux
Hardware	Standard CPU Workspace

8 Hyperparameter Selection

Parameter	Value
KNN 'n_neighbors'	5 (default)
Naïve Bayes 'alpha'	1.0 (default Laplace smoothing)

9 Results

Metric	Value
KNN Accuracy	0.9022
KNN Precision	0.8930
KNN Recall	0.8523
KNN F1-score	0.8722
MNB Accuracy	0.7951
MNB F1-score	0.7516

10 Result Visualization

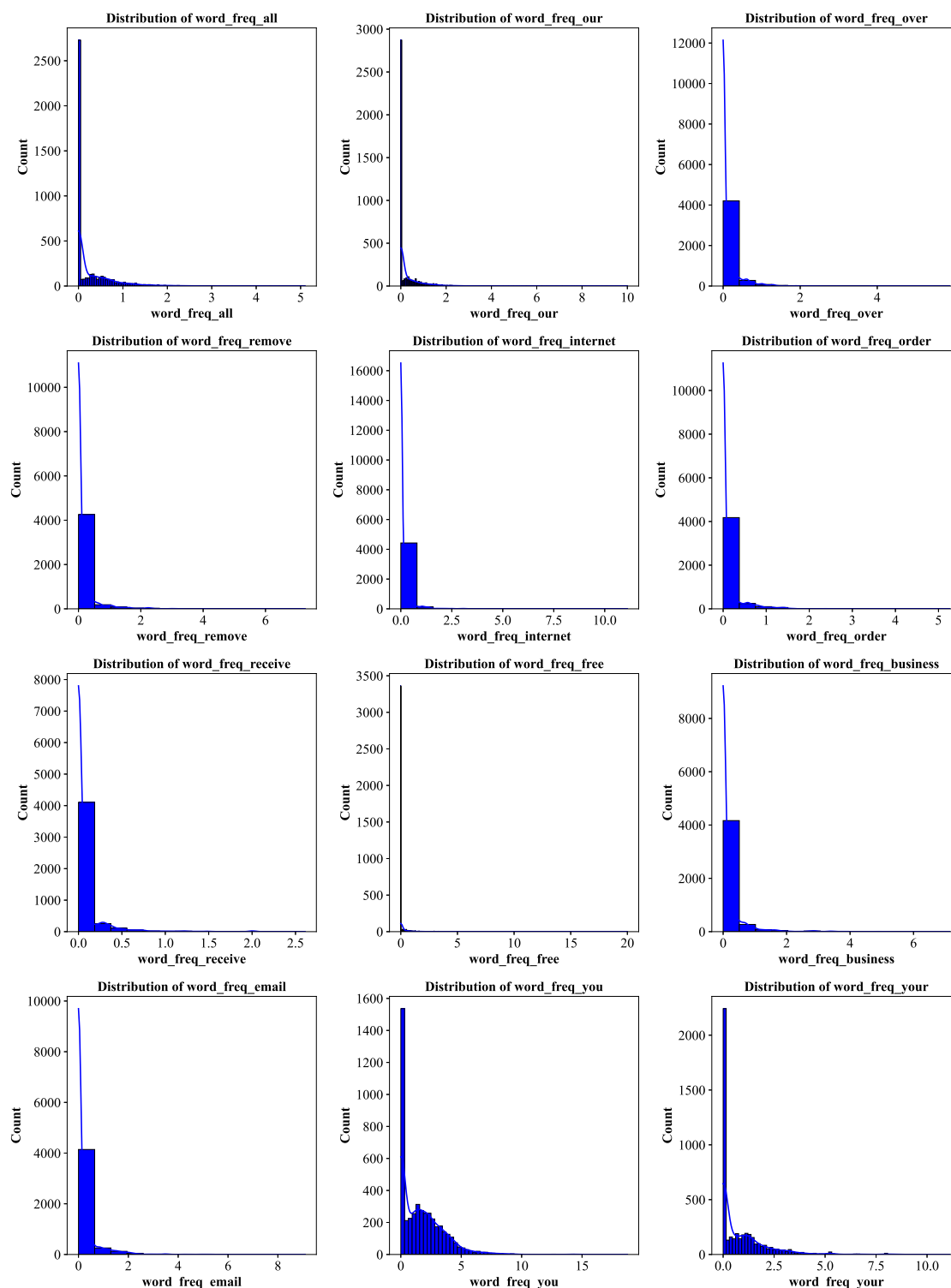


Figure 1: Consolidated EDA summary showcasing the distributions of top predictive features within the Spambase dataset.

Inference: A lot of these features are right-skewed, which honestly isn't surprising for word-frequency data – most emails just don't contain a given word very often. That skew

is part of why plain Naïve Bayes tends to struggle here compared to a method like KNN that doesn't assume anything about the shape of the distribution.

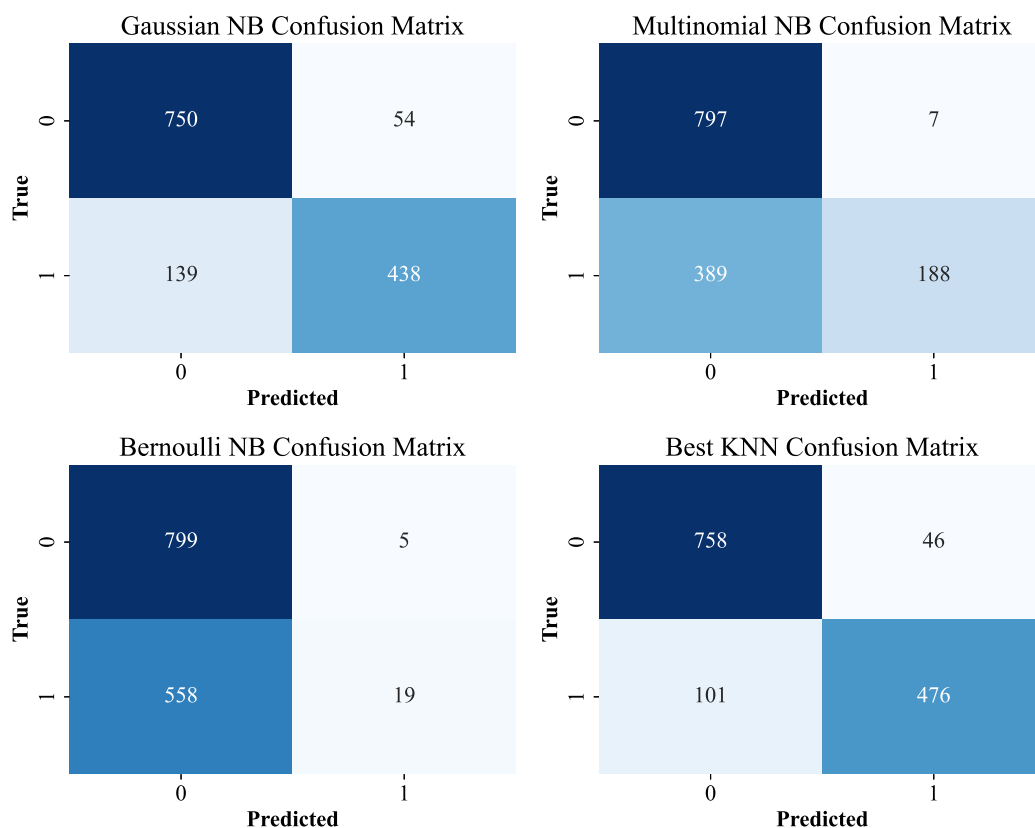


Figure 2: Confusion Matrix comparison between Naïve Bayes and K-Nearest Neighbors.

Inference: Looking at the two confusion matrices side by side, KNN gets more true positives and true negatives than Naïve Bayes does. Naïve Bayes in particular tends to flag a fair number of legitimate emails as spam, which shows up as a higher false positive count.

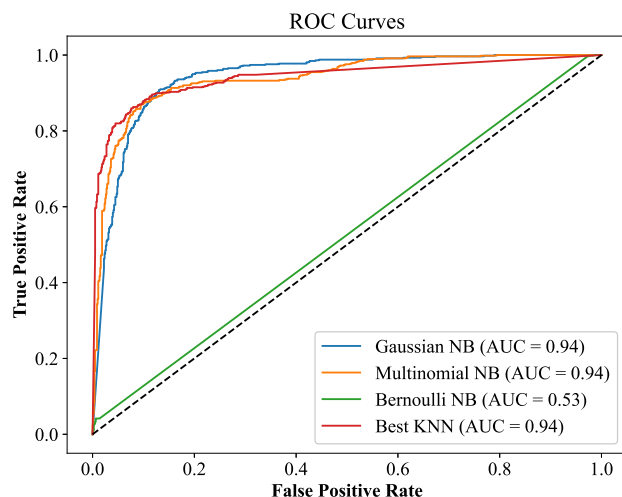


Figure 3: Receiver Operating Characteristic (ROC) curves demonstrating KNN’s superior Area Under the Curve (AUC).

Inference: KNN’s ROC curve sits above Naïve Bayes’s for nearly the entire range of thresholds – roughly 0.94 AUC versus 0.84. So across the board, not just at one particular cutoff, KNN comes out ahead here.

11 Discussion

KNN was the clear winner in this comparison – about 90% accuracy against MNB’s 79%, with a similar gap in F1-score. My guess is this comes down to the independence assumption Naïve Bayes relies on. In an email, word choices aren’t independent at all; certain words tend to show up together (think “free,” “win,” “click” all appearing in the same spam message), and that correlation is exactly what the naïve assumption throws away.

12 Conclusion

Working through this lab gave a decent side-by-side look at a probabilistic method and a distance-based one. Once the features were properly scaled, KNN did a solid job separating spam from ham, better than Naïve Bayes managed on the same data. Plots were saved in EPS format to keep things consistent with the lab’s reporting requirements.

13 References

1. Cover, T., and Hart, P. “Nearest neighbor pattern classification.” *IEEE Transactions on Information Theory*, 1967.
2. McCallum, A., and Nigam, K. “A comparison of event models for naive bayes text classification.” *AAAI-98 workshop on learning for text categorization*, 1998.